Ammara Talib
Website: https://ammara-talib.github.io/ *Email:* ammara.talib@.edu , Google Scholar

RELEVANT EXPERIENCE

Dr. Ammara Talib is an Environmental scientist and engineer who is interested in carbon and water fluxes and in understanding how climate change, as well as disturbances, may impact society through changes in the patterns of water availability. Some highlights of Dr. Talib's scientific career include:

- Developed and applied sophisticated process-based (SWAT, HSPF) and data-driven models (Random Forest, XGBoost, LSTM/CNN) that are used for the prediction and evaluation of water fluxes in different types of ecosystems (Talib et al, 2021, Talib et al., 2023, Talib et al., 2024)
- Corrected an irrigation water management tool (WISP) that is designed to help growers optimize crop water use efficiently (Talib et al., 2024)
- Carried out fieldwork in a state-of-the-art experiment about water use by potatoes and pine in Wisconsin
- Awarded Fulbright scholarship (2012-2015), Chancellor's Opportunity Award (2015), METER, CPEP Seed Grant Award Center of Climate Research (2017), METER Grant A Harris instrumentation fellowship (2019)

EDUCATION

University of Wisconsin Madison **Aug 2023**
Ph.D. Civil & Environmental Engineering (Specialization in Water resource engineering)

Dissertation: Prediction and forecasting of Evapotranspiration and Groundwater anomalies, along with improved parameterization of ET in agricultural lands
Advisor: Dr. Ankur Desai, Ecometeorology Lab

University of Massachusetts Amherst **June 2015**
MS Environmental Conservation (Specialization in Water, Watersheds, and Wetlands)

Master's Thesis: Assessment of impacts of land use land cover (LULC) and climate change on water resources in SuAsCo watershed, Massachusetts
Advisor: Dr. Timothy Randhir, Ecowater Lab

University of Punjab Lahore Pakistan
BS Earth & Environmental Sciences

Undergraduate Thesis: Environmental Impact Assessment of Pan Power plant
Advisor: Prof. Muhammad Jamal-Ud-Din Ahmed

RESEARCH INTERESTS

Measurements of hydrologic cycle and carbon cycle components and forecast models of complex ecosystems involving: Tower mounted eddy covariance, Satellite measurements, analysis of land-atmosphere interactions, process-based models, statistical and data driven deep learning models, climate change and landuse impacts

RESEARCH EXPERIENCE

Yale School of Environment, Yale University **Oct 2024 – Aug 2025**
Postdoctoral Research Associate *New Haven, Connecticut, USA*

- How water use efficiency changes within ecosystems and whether differences in WUE across ecosystems resemble ecosystem variability
- Determine if WUE in coastal ecosystems denotes salinity stress.

Harvard Forest, Harvard University **Sep 2023 – Sep 2024**
Postdoctoral Research Associate *Petersham, Massachusetts, USA*

- Analysis of temporal and spatial variability in precipitation and streamflow to evaluate the impacts of climate change on eastern USA Forests
- Quantify the role of ecosystem disturbance and succession on vegetation-climate and water coupling.

Department of Civil & Environmental Engineering, **Sep 2017 - Aug 2023**
UW Madison
Research Assistant, with Prof. Ankur Desai. *Madison, Wisconsin USA*

- Developing a Hydrologic model for water budget in Central Sands Wisconsin
- Improving parameterization of a regional irrigation scheduling program for evapotranspiration estimation with eddy covariance measurements

Department of Civil & Environmental Engineering, **Sep 2017 - Aug 2023**
UW Madison
Research Assistant, with Prof. Ankur Desai. *Madison, Wisconsin USA*
- Developing a Hydrologic model for water budget in Central Sands Wisconsin
- Improving parameterization of a regional irrigation scheduling program for evapotranspiration estimation with eddy covariance measurements
- Evaluation of prediction and forecasting models for evapotranspiration of agricultural lands in the Midwest U.S.
- Spatial and Temporal Forecasting of Groundwater anomalies

Environmental Fluid Mechanics Lab, Department of Civil & Environmental Engineering, **Sep 2015 – Sep2016**
UW Madison
Research Assistant, with Prof. Chin, Wu *Madison, Wisconsin USA*
- Prediction modeling for nitrogen and phosphorus in agricultural watersheds

Department of Environmental Conservation **Sep 2012 - Jun 2015**
University of Massachusetts Amherst, USA
Fulbright research fellow, with Prof. Timothy Randhir *Madison, Wisconsin USA*
- Assessment of impacts of land use land cover (LULC) and climate change on water resources in SuAsCo watershed, Massachusetts
- Managing emerging contaminants in watersheds: need for comprehensive, systems-based strategies

Department of Earth & Environmental Sciences **Jan 2010-Jun 2012**
University of Punjab Lahore
Research Assistant, with Prof. Muhammad Jamal-Ud-Din Ahmed *Lahore, Pakistan*
- Environmental Impact Assessment of Pan power plant near residential area in Pakistan

PUBLICATIONS **_Peer reviewed_**

Talib, A., *Desai, A. R., & Huang, J.* ***(2024).*** Spatial and temporal forecasting of groundwater anomalies in complex aquifer undergoing climate and land use change**.** Journal of Hydrology**,** https://doi.org/10.1016/j.jhydrol.2024.131525

Talib, A., *Desai, A. R., Huang, J., (**2024**).* Improving parameterization of a regional irrigation scheduling program for evapotranspiration estimation with eddy covariance measurements. Agricultural and Forest Meteorology, https://doi.org/10.1016/j.agrformet.2024.109967

Talib, A., *& Randhir, T. O.* ***(2023).*** Long-term effects of land-use change on water resources in urbanizing watersheds. PLOS Water, 2(4). https://doi.org/10.1371/journal.pwat.0000083

Desai, A.R., J. Thom, S. Wiesner, B. Butterworth, N. Koupaei-Abyazani, A. Merrelli, B. Murphy, A. Muttaqin, S. Paleri, ***A. Talib****, J. Turner, J. Mineau**.*** ***(2022).*** Drivers of decadal carbon fluxes across temperate ecosystems. Journal of Geophysical Research: Biogeosciences, https://doi.org/10.1029/2022JG007014

Ebert, L. ***A., Talib,*** *A., Zipper, S. C., Desai, A. R., Paw U, K. T., Chisholm, A. J., Prater, J., & Nocco, M. A. (**2022**).* How High to Fly? Mapping Evapotranspiration from Remotely Piloted Aircrafts at Different Elevations. Remote Sensing, 14(7). https://doi.org/10.3390/rs14071660

Talib, A., *Desai, A. R., Huang, J., Griffis, T. J., Reed, D. E., & Chen, J.* ***(2021).*** Evaluation of prediction and forecasting models for evapotranspiration of agricultural lands in the Midwest US. Journal of Hydrology, 126579. https://doi.org/10.1016/j.jhydrol.2021.126579

Chatterjee, S., Stoy, P. C., Debnath, M., Nayak, A. K., Swain, C. K., Tripathi, R., Chatterjee, D., Mahapatra, S. S., ***Talib, A****., & Pathak, H.* ***(2021).*** Actual evapotranspiration and crop coefficients for tropical lowland rice (Oryza sativa L.) in eastern India. Theoretical and Applied Climatology, 146(1–2). https://doi.org/10.1007/s00704-021-03710-0

Talib, A *and T. O. Randhir.* ***(2017).*** Climate change and land use impacts on hydrologic processes of watershed systems Journal of Water and Climate Change https://doi.org/10.2166/wcc.2017.064

Talib, A. *and T. O. Randhir* ***(2017****),* Managing emerging contaminants in watersheds: Need for comprehensive, systems-based strategies, Sustainability of Water Quality and Ecology, doi: https://doi.org/10.1016/j.swaqe.2016.05.002.

Talib, A. *and Randhir T. O.* ***(2016)*** Managing emerging contaminants: status, impacts, and watershed-wide strategies. Exposure and Health 8:143–158. doi: 10.1007/s12403-015-0192-4

In Preparation

Talib A. *Matthes J, Green M,* Analysis of Increasing Evapotranspiration Across New England Long-Term Forest Research Sites. (In preparation)

Talib A. *Matthes J,* Coupled Carbon and Water Cycling in a Temperate Forest Under Increasing Climatic Disturbance. *(In preparation)*

Orwig, D., Matthes, J., ***Talib, A****., Boose E., Munge, J.W., Chung, K., Inyama, F., et al.,* Long-

Term Shifts in Forest Structure and Ecosystem Function with the Decline of a Foundation Species, Global Change Biology. (In preparation)

INVITED PRESENTATIONS

- *Desai, A.R.,* ***Talib A****,* Water use by crops and forests in the Central Sands. Wisconsin Potato Vegetables & Growers Association, Grower Education Conference & Industry Show, Stevens Point, WI ***Feb 7, 2023***
- *Desai, A.R.,* ***Talib A****,* Predicting Central Sands crop water use. Wisconsin Potato Vegetables & Growers Association, Grower Education Conference & Industry Show,Stevens Point, WI ***Feb 9, 2022***
- ***Talib A****, Desai, A. R. (2020)* What have we learned from continuous crop and forest evapotranspiration observations in the Central Sands? Wisconsin Potato Vegetables & Growers Association, Grower Education Conference & Industry Show, Stevens Point, WI ***Feb 5, 2020***
- ***Talib A****, Desai, A. R. (2019)* Improving forecasts of crop water demand with direct ET measurements over irrigated fields, Wisconsin Potato Vegetables & Growers Association, Grower Education Conference Industry Show Stevens ***Feb 5, 2019***
- ***Talib, A.*** *and Desai, A. R. 2018,* Water from ground to sky: New approaches to observing and predicting field to basin scale ET over crops and plantations, Wisconsin Potato Vegetables & Growers Association, Grower Education Conference & Industry Show, Stevens Point ***Feb 6, 2018***

Oral PRESENTATIONS

- ***Talib A****, Matthes J,* Examining Interactions of Disturbance, and Water Cycling in the Eastern US with Ecosystem-Atmosphere Observations and Models, Saint Paul, MN, Water Science Conference, 2024. ***24-27 Jun, 2024***
- *Desai, A.R., J. Thom, S. Wiesner, B. Butterworth, N. Koupaei-Abyazani, A. Merrelli, B. Murphy, A. Muttaqin, S. Paleri,* ***Talib A.****, J. Turner, J. Mineau. (2021)* From half-hour to quarter century- Drivers of carbon fluxes across a northern ecosystem tower cluster**,** AGU Fall Meeting ***13-17 Dec, 2021***
- ***Talib A****, Desai, A. R. (2020)* Water Use by Crops and Forest and Improving Irrigation Planning- American Water Resource Association (AWRA) 44th Annual Conference ***12 March, 2020***
- ***Talib A****, Desai, A. R. (2020)* Improving irrigation planning and early prediction for agricultural drought in Wisconsin, AGU Fall Meeting ***07-11 Dec , 2020***
- *LA Ebert, AJ Chisholm, J Prater, SC Zipper,* ***Talib A****, AR Desai, MA Nocco. (2020)* How high to fly? Evaluating different elevations for mapping evapotranspiration from remotely piloted aircrafts, American Geophysical Union (AGU) Fall Meeting ***07-11 Dec, 2020***
- ***Talib A****, Desai, A. R. (2019)* Field-scale mapping and forecasting of water budgets in intensively irrigated agricultural regions through an advanced ensemble modeling framework, American Geophysical Union (AGU), Fall Meeting ***09-13 Dec, 2019***
- *LA Ebert, AJ Chisholm, J Prater, SC Zipper,* ***Talib A****, AR Desai, MA Nocco. (2019)* Using remotely piloted aircrafts to evaluate potato water stress in Central Wisconsin**,** American Geophysical Union (AGU), Fall Meeting ***09-13 Dec, 2019***

- ***Talib A**, Matthes J,* Examining Interactions of Disturbance, and Water Cycling in the Eastern US with Ecosystem-Atmosphere Observations and Models, Saint Paul, MN, Water Science Conference, 2024. ***24-27 Jun, 2024***

- *Desai, A.R., J. Thom, S. Wiesner, B. Butterworth, N. Koupaei-Abyazani, A. Merrelli, B. Murphy, A. Muttaqin, S. Paleri, **Talib A.**, J. Turner, J. Mineau. (2021)* From half-hour to quarter century- Drivers of carbon fluxes across a northern ecosystem tower cluster**,** AGU Fall Meeting ***13-17 Dec, 2021***

- ***Talib A**, Desai, A. R. (2020)* Water Use by Crops and Forest and Improving Irrigation Planning- American Water Resource Association (AWRA) 44th Annual Conference ***12 March, 2020***

- ***Talib A**, Desai, A. R. (2020)* Improving irrigation planning and early prediction for agricultural drought in Wisconsin, AGU Fall Meeting ***07-11 Dec , 2020***

- *LA Ebert, AJ Chisholm, J Prater, SC Zipper, **Talib A**, AR Desai, MA Nocco. (2020)* How high to fly? Evaluating different elevations for mapping evapotranspiration from remotely piloted aircrafts, American Geophysical Union (AGU) Fall Meeting ***07-11 Dec, 2020***

- ***Talib A**, Desai, A. R. (2019)* Field-scale mapping and forecasting of water budgets in intensively irrigated agricultural regions through an advanced ensemble modeling framework, American Geophysical Union (AGU), Fall Meeting ***09-13 Dec, 2019***

- *LA Ebert, AJ Chisholm, J Prater, SC Zipper, **Talib A**, AR Desai, MA Nocco. (2019)* Using remotely piloted aircrafts to evaluate potato water stress in Central Wisconsin**,** American Geophysical Union (AGU), Fall Meeting ***09-13 Dec, 2019***

- ***Talib, A**. and Desai, A. R; (2018).* Efficacy of Machine Learning Algorithms for Identifying Hotspots of Groundwater Depletion in Intensively Irrigated Agricultural Regions, American Geophysical Union (AGU), Fall Meeting ***10-14 Dec 2018***

- *Nocco, M. A. ; McNamee, E. O. ; Bohman, B. ; **Talib**, A. ; Rosen, C. J. ; Desai, A. R. ; Kern, C. C. ; Kucharik, C. J. ; Twine, T. E. (2018)* Pine to Potato Conversion Impacts to Groundwater Recharge in the Northern Great Lakes Region, American Geophysical Union (AGU) ***10-14 Dec 2018***

- ***Talib, A.** and Desai, A. R. 2017.* Groundwater-Surface water interaction in agricultural watershed that encompasses dense network of High-Capacity wells, American Geophysical Union (AGU), Fall Meeting ***11-15 Dec, 2017***

- ***Talib, A** and Randhir. 2014.* Land Use Land Cover Impacts on Water Quantity and quality in Watershed Systems, Lombard, Illinois, 69th SWCS International Annual Conference, making waves in conservation, our life on land and its impacts on water ***27-30 July, 2014***

- ***Talib, A** and Randhir. 2014.* Integrated Assessment of Land Use Land Cover and Climate Change Impacts on Water Quantity and Quality in SuAsCo Watershed System, 2014 American Water Resource Association (AWRA) Summer Specialty Conference on Integrated Water Resources Management (IWRM) from theory to application Reno, Nevada ***1-3 July 2014***

- ***Talib, A** and Randhir. 2014.* Climate Change Impacts on Water Quantity and Quality in Watershed Systems, UCOWR-NIWR-CUAHSI Conference on Water Systems, Science, and Society under Global Change, at Tuft University Medford, Massachusetts ***18-20 June, 2014***

- ***Talib A**, and Randhir. 2014.* Climate Change Impacts on Water Resources, Northeast Natural History Conference (NENHC), Springfield *Massachusetts* ***7-9 April 2014***

Academic Awards and fellowship

AWARDS AND FELLOWSHIPS

- UW Water Resources Institute, Data-driven groundwater depth and risk forecasting in the Central Sands region of WI for sustainable management (I contributed to writing the proposal, then funded by it as research assistant on the grant), $71,499 ***July 2021—Dec 2022***
- State of Wisconsin Dept of Natural Resources, Evapotranspiration of irrigated crops: Monitoring and data collection (I contributed to writing the proposal, then funded as research assistant on the grant), $15,000 ***June 2018- Dec 2018***
- Wisconsin Potato and Vegetation Growers Association (WPVGA), Monitoring evapotranspiration in Central Sands farms and forests, (I contributed to writing proposal, then funded as research assistant on the grant) $117,012 ***July 2018-June 2023***
- METER Grant A Harris instrumentation fellowship, $10,000 ***March 2019***
- CPEP Seed Grant Award Center of Climate Research, Observing Wisconsin Central Sands Water Budget Component Under High Groundwater Demand and a Changing Climate, $8000 ***May 2017-Aug 2017***
- Chancellor's Opportunity Award, University of Wisconsin Madison $3000 ***Sep 2015***
- Academic scholarship, University of Massachusetts Amherst $300 ***April 2015***
- Fulbright Scholarship (University of Massachusetts Amherst, USA) $980,00 ***Sep 2012- May 2015***
- Merit scholarship for two undergraduate semesters (College of Earth & Environmental Sciences in Punjab University Lahore, Pakistan), $200 ***March 2007-May 2009***
- Honors Rank student (Islamic College Cooper Road, Lahore, Pakistan ***May 2004***

WORKSHOPS AND PROFESSIONAL DEVELOPMENT

CERTIFICATES

- Neural Networks and Deep Learning, Coursera ***July 2018***
- Improving Deep Neural Networks- Hyperparameter Tuning, Regularization and Optimization, Coursera ***July 2018***
- Structuring Machine Learning Projects, Coursera ***July 2018***
- Convolutional Neural Networks, Coursera ***Aug 2018***
- Sequence Models, Coursera ***Aug 2018***

WORKSHOPS

- Graduate Assistants' Equity Workshops, Division of Diversity, Equity & Educational Achievement ***Jan 2016***
- Graduate Assistants' Equity Workshops, Division of Diversity, Equity & Educational Achievement ***Sep 2016***

TEACHING AND MENTORING

- Mentor, Undergraduate summer research program, Department of atmospheric and oceanic sciences, University of Wisconsin Madison ***June-Aug 2022***
- Teaching Assistant for Introductory Biology course (Bio 151, Bio 152). University of Wisconsin Madison ***Jan 2018-May 2023***
- Teaching Assistant for Meteorological measurement course (AOS 404). University of Wisconsin Madison ***Sep 2018-Dec 2018***
- Science Teacher Quest High school, Lahore, Pakistan ***Sep 2006-Aug 2011***

PROFESSIONAL AND LEADERSHIP ROLE

- Member at "Outreach committee at water resource engineering (WRE) ***Jan 2021-May 2023***
- Member at "Cardiac on campus" organization ***Jan 2021-May 2023***
- Volunteer intern at Waste busters Pakistan Lahore ***Jun-Aug 2011***
- Volunteer intern at Environmental Protection Agency (EPA), Pakistan ***Jun-Aug 2010***
- Volunteer intern at Greenlays clean Pakistan ***Jun-Dec 2009***

VOLUNTEER AND COMMUNITY SERVICE

- Volunteer in the Homeless shelter (Christian Community Action) ***Oct-Present 2024***
- Volunteer in the Connecticut Hospice, Inc. ***Nov-Present 2024***
- Volunteer in the Cancer unit in Massachusetts General Hospital ***April-Aug 2024***
- Volunteer in the Neuroscience/stroke department at UW University of Wisconsin Hospital and clinics ***Jan-Jun 2023***
- Caretaker for a quadriplegic patient ***Jan-Aug 2023***
- Volunteer Crisis Counselor at Crisis Text Line ***Aug 2021-Present***
- Volunteer Listener for emotional support at 7 cups emotional health service and online therapy provider ***Dec 2020-Present***
- Volunteer in the General Surgery department at UW University of Wisconsin Hospital and clinics ***Nov-Dec 2022***
- Volunteer in the Emergency Room (ER) at UW health ***Sep 2019-Mar 2020***
- Caretaker/Home Health Aide for Patients with Parkinson and cerebral Palsy ***July-Sep 2016***
